Dual-Path Self-Supervised Spatiotemporal Learning for Satellite Video Scene Understanding

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***Abstract*—Satellite video scene understanding (SVSU) is critical for dynamic Earth observation but faces challenges from the distinctive spatiotemporal patterns and the limited publicly available data. While self-supervised learning (SSL) has advanced representation learning for remote sensing (RS) imagery, its application to SVSU remains challenging due to the need for temporal motion modeling and data scarcity. To address these limitations, we propose a novel dual-path self-supervised spatiotemporal learning framework that disentangles spatial and temporal representation learning. A frozen vision transformer (ViT) backbone, pretrained on large-scale RS images, preserves robust spatial semantics, whereas a trainable temporal encoder is introduced to capture motion dynamics specific to satellite videos. A feature interaction adapter (FIA) is integrated to fuse the information from both paths. To empower temporal modeling, we devise a dual reconstruction objective that jointly optimizes masked spatial appearance reconstruction and dynamic motion reconstruction during self-supervised pretraining. The experimental results demonstrate the superior performance of the proposed method.**

***Index Terms*—Satellite video, scene understanding, self-supervised learning (SSL), spatiotemporal representation.**

I. Introduction

THE rapid development of advanced satellite constellations has enabled persistent, high-resolution video observation across large terrestrial and maritime regions. This continuous monitoring supports new possibilities for dynamic Earth surface analysis [1], [2], [3], [4]. Applications such as traffic management, environmental monitoring, and maritime surveillance increasingly depend on satellite video scene understanding (SVSU) [5], [6], [7], namely the extraction and interpretation of complex spatiotemporal patterns in satellite video data.

Self-supervised learning (SSL) [8], [9], [10], [11] has emerged as a dominant paradigm in computer vision, driving substantial progress in remote sensing (RS) across various data modalities [12], [13], [14]. However, this success has predominantly been confined to static image data. A pronounced imbalance persists in data availability: while a wealth of publicly accessible datasets supports robust pretraining for RS imagery, RS video datasets remain limited in both size and diversity. This scarcity not only hinders the training of large, generalizable models for SVSU but also increases the risk of overfitting.

A common strategy to mitigate this data bottleneck involves knowledge transfer [15], [16] from models pretrained on abundant RS images. However, this approach faces two critical problems in the context of SVSU. First, beyond the understanding of static spatial information, a crucial aspect of SVSU is the ability to capture sparse and typically small motion cues within the video scene. Existing spatial-only backbones, primarily trained on images, inherently lack the capability to effectively model these temporal dynamics. Second, with only limited satellite video available, fine-tuning or SSL often provides an insufficient training signal and tends to erode the transferred spatial priors, reducing performance. This raises a central research question: how can we leverage the rich spatial knowledge learned from large-scale RS image datasets, while simultaneously learning temporal motion cues for SVSU, without eroding spatial priors under data scarcity?

To address this issue, we propose a dual-path self-supervised spatiotemporal learning framework. The framework separates spatial and temporal representation learning, both of which are essential for scene understanding. The spatial path employs a vision transformer (ViT) [17] backbone pretrained on large-scale RS images. Crucially, this spatial path remains frozen throughout training and inference to preserve its deeply learned semantic structures, providing contextual understanding. Concurrently, a temporal encoder is introduced and made trainable, specifically designed to capture the motion dynamics inherent in satellite videos. The two paths are fused by a feature interaction adapter (FIA) to produce spatiotemporal features for downstream scene understanding tasks.

To empower the temporal path to learn spatiotemporal features, our SSL strategy integrates a dual reconstruction objective: in addition to reconstructing masked spatial appearance, the model also reconstructs motion information from video frames. This objective drives the temporal path to learn motion-sensitive representations without eroding the spatial priors of the frozen backbone, yielding features that integrate both spatial context and temporal motion for effective scene understanding.

In summary, the main contributions of this article are as follows.

1) A tailored SSL scheme with a dual reconstruction objective is designed, where both masked appearance and temporal motion dynamics are leveraged to obtain discriminative spatiotemporal representations.

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2) An SVSU-oriented framework designed for satellite video characteristics is proposed, in which large-scale image-pretrained spatial representations are preserved while complementary temporal dynamics are simultaneously learned, enabling robust spatiotemporal feature extraction in settings with limited satellite video data.
3) Comprehensive experimental evaluations are conducted, through which the superiority of the proposed method is demonstrated, establishing an effective and scalable paradigm for SVSU.

The remainder of this article is organized as follows. Section II reviews the related work in satellite video analysis and SSL. Section III details the proposed framework. Section IV presents the experimental results and analysis. Finally, Section V concludes this article.

II. Related Works

A. Advances in Satellite Video Analysis

Satellite video analysis has attracted increasing attention in recent years. A primary focus within this domain has been the challenging task of moving target detection and tracking. Yin et al. [18] presented a dataset specifically designed for detecting and tracking small and dense moving objects in satellite videos, providing crucial baseline results that serve to guide future research. DSFNet [19] introduced a fusion approach, integrating 2-D and 3-D backbone networks to extract both static contextual cues and dynamic motion features. Pan et al. [20] proposed LSCNet, a lightweight and efficient detection network that introduced a plug-and-play object-cluster module and a multiframe feature-aggregation module. More recently, advanced tracking methods have emerged, such as JSANet proposed by Song et al. [21], which leverages self-attention for feature refinement and cross-attention for modeling interdependencies between template and search branches, leading to more robust tracking. Zhang et al. [22] developed a tracker that explicitly learns motion and background information through relational modeling, enhancing feature discriminability and better distinguishing targets from distractors.

Beyond these low-level tasks focused on individual moving targets, increasing attention has been paid to higher-level scene understanding in satellite videos. These include recognizing complex events and classifying scene types, covering tasks such as satellite video scene classification (SVSC) and event of interest (EOI) detection. Early efforts in SVSC, pioneered by Gu et al. [5], combined spatial and motion features within a two-stage CNN–LSTM framework. Subsequently, Wang et al. [6] proposed a low-rank sparse representation two-stream network (LSRTN), which explicitly decomposed videos into low-rank background and sparse motion components, feeding them into a dual CNN pipeline to enhance classification accuracy. These supervised designs demonstrate the effectiveness of separating background and motion for satellite video understanding. In contrast, we adopt a decomposition-based strategy to construct a self-supervised motion reconstruction target, enabling motion-sensitive representation learning for SVSU under limited labeled data and satellite video availability.

More recently, to address challenges, such as data scarcity and multilabel scenarios, Guo et al. [7] curated a dataset with 18 distinct scene categories and introduced an STFCE network. Complementing scene classification, significant efforts have also been directed toward EOI detection in satellite videos. For instance, Gu et al. [23] developed a two-stream pipeline to identify meaningful events, utilizing an AlexNet-based branch for static scene context analysis and a trajectory-analysis branch to capture motion consistency. Furthermore, Wang et al. [24] proposed a deep metric learning framework that integrates spatiotemporal slice analysis, thereby achieving more robust detection performance.

B. SSL in RS

SSL has rapidly emerged as a dominant paradigm in computer vision, achieving breakthrough performance for both image and video representation learning. Early SSL studies focused on designing pretext tasks [25], [26], [27], [28], [29], such as jigsaw puzzle solving, rotation prediction, and temporal order verification, to extract semantic structure from unlabeled visual data. With the advent of contrastive learning frameworks [30], [31], [32], SSL began to surpass supervised learning in transferability, a trend further accelerated by the rise of generative masking approaches [33], [34], [35], [36], [37], such as MAE [38], BEiT [39], and VideoMAE [40], which exploit the reconstruction of masked tokens to drive efficient representation learning. While VideoMAE [40] focuses on generic video representation learning, our work emphasizes a dual-path design tailored to satellite videos under limited data. MotionMAE [41] jointly models appearance and motion via frame-difference reconstruction, and DiST [42] adopts a dual-encoder design for generic video settings. In contrast, our work is formulated as a self-supervised framework for data-scarce satellite videos, where motion supervision is derived from low-rank sparse decomposition and RS spatial priors are preserved via a frozen pretrained spatial encoder.

In the RS domain, SSL has become particularly attractive, primarily due to the high costs and specialized expertise required for large-scale annotation. Sun et al. [43] proposed RingMo, a generative self-supervised RS foundation model that utilizes a large-scale dataset of two million satellite and aerial images and a masked image modeling strategy specifically designed for RS data, achieving state-of-the-art performance across multiple downstream tasks. In addition, Liu et al. [44] introduced RemoteCLIP, a vision–language RS foundation model leveraging unified large-scale image–caption datasets through annotation unification and data scaling, achieving strong results in cross-modal retrieval and zero-shot classification. Furthermore, Dong et al. [45] presented a generative ConvNet foundation model for RS image interpretation, incorporating sparse modeling and low-frequency reconstruction. For RS image time series, Yao et al. [46] proposed RingMo-Sense, a foundation model designed for spatiotemporal prediction via spatiotemporal evolution disentangling, achieving superior performance across a range of downstream RS tasks. In addition, Tarasiou et al. [47]

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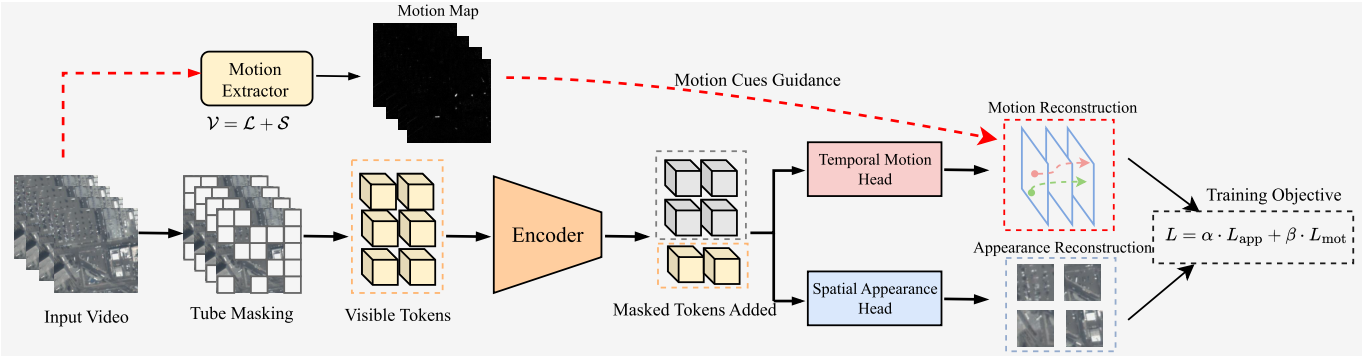


Fig. 1. Overview of the proposed framework. The input video is first partitioned into tube tokens with partial masking, and then processed by an encoder. A motion extractor provides additional motion cues to guide reconstruction. Two decoder heads are employed to recover temporal motion and spatial appearance, respectively, and the joint training objective integrates both components.

introduced TSViT, a fully attentional temporal–spatial ViT architecture tailored for general satellite image time-series processing, which achieves state-of-the-art results in semantic segmentation and classification.

Despite these notable advancements, SSL approaches specifically designed for SVSU remain underexplored. This research gap primarily stems from two critical obstacles: the scarcity of large-scale, publicly available satellite video datasets and the inherent difficulty of learning robust motion representations within satellite videos.

III. Proposed Method

This section details the proposed dual-path self-supervised spatiotemporal learning framework for satellite video understanding. As illustrated in Fig. 1, the framework is designed to learn robust spatiotemporal representations by jointly integrating spatial appearance and motion cues. The architecture of the proposed dual-path spatiotemporal encoder is shown in Fig. 2. To fully exploit the rich knowledge obtained from large-scale RS image datasets pretraining and to address the scarcity of publicly available satellite video datasets, the framework adopts a dual-path design. Specifically, a frozen spatial backbone based on a pretrained ViT preserves strong spatial semantics learned from extensive RS imagery, while a trainable temporal encoder path is introduced to capture dynamic cues in satellite videos. This design safeguards spatial priors in data-limited regimes and simultaneously strengthens temporal motion modeling.

A. SSL With Dual Reconstruction

In order to capture both static appearance information and dynamic motion cues from satellite videos, we design a dual reconstruction SSL strategy. As illustrated in Fig. 1, the framework employs two complementary reconstruction branches: one branch aims to recover the spatial appearance of masked patches, thereby preserving scene semantics, while the other branch focuses on reconstructing temporal dynamics derived from satellite videos to model motion patterns.

Let $\mathcal{V}$ be an unlabeled collection of satellite video clips, where each video is represented as a 4-D tensor $V \in \mathbb{R}^{T\times H\times W\times 3}$ consisting of T frames with spatial resolution $H \times W$. Each video clip is divided into nonoverlapping spatial patches of size $P \times P$ pixels while retaining the temporal dimension. This produces $N = HW/P^2$ spatial locations for each frame. After linear projection into a d-dimensional embedding space, the patch tokens from all T frames are concatenated to form the input sequence $\mathbf{X} \in \mathbb{R}^{T\times N\times d}$. Due to the highly correlated static background in satellite videos, random masking may lead to information leakage, thereby reducing the effectiveness of representation learning. To address this issue, we employ a tube masking strategy during self-supervised pretraining, following VideoMAE [40]. In this scheme, patches at the same spatial location are either entirely masked or entirely visible across all frames, which preserves temporal consistency and facilitates more effective temporal modeling.

A binary mask $\mathbf{M}$ is sampled from a predefined masking distribution $\mathcal{P}_{\text{mask}}$, indicating which tube-patches are masked. Let Ω be the set of masked patch indices and $\bar{\Omega}$ the set of visible indices. The encoder f_θ processes only the visible tokens $\mathbf{X}_{\bar{\Omega}}$ to produce the latent representation

$$\mathbf{Z} = f_\theta(\mathbf{X}_{\bar{\Omega}}). \tag{1}$$

The latent features $\mathbf{Z}$, together with learnable mask tokens $\mathbf{z}^{\text{mask}}$ reinserted at the masked positions, are fed into two lightweight transformer decoders: an appearance decoder g_ϕ^{app} and a motion decoder g_ψ^{mot}. The appearance decoder is responsible for reconstructing spatial semantics, while the motion decoder explicitly captures dynamic patterns. Formally, the reconstruction process is defined as follows:

$$\hat{\mathbf{X}}^{\text{app}} = g_\phi^{\text{app}}(\mathbf{Z}, \mathbf{z}^{\text{mask}}), \qquad \hat{\mathbf{X}}^{\text{mot}} = g_\psi^{\text{mot}}(\mathbf{Z}, \mathbf{z}^{\text{mask}}) \tag{2}$$

where $\hat{\mathbf{X}}^{\text{app}}$ and $\hat{\mathbf{X}}^{\text{mot}}$ denote the reconstructed appearance and motion signals, respectively.

The appearance reconstruction branch focuses on recovering the precise pixel-level content of each masked tube-patch. For every spatiotemporal location $(t, i) \in \Omega$, let $\mathbf{p}_{t,i}$ be the ground-truth RGB pixel values, and $\hat{\mathbf{p}}_{t,i}$ be the corresponding predicted RGB values from the appearance decoder. To quantify the discrepancy between the reconstructed and ground-truth appearance, we employ a mean squared error (mse) loss, which is widely adopted for pixel-level reconstruction tasks due to its effectiveness in promoting visual fidelity

$$\mathcal{L}_{\text{app}} = \frac{1}{|\Omega|} \sum_{(t,i)\in\Omega} \|\mathbf{p}_{t,i} - \hat{\mathbf{p}}_{t,i}\|_2^2. \tag{3}$$

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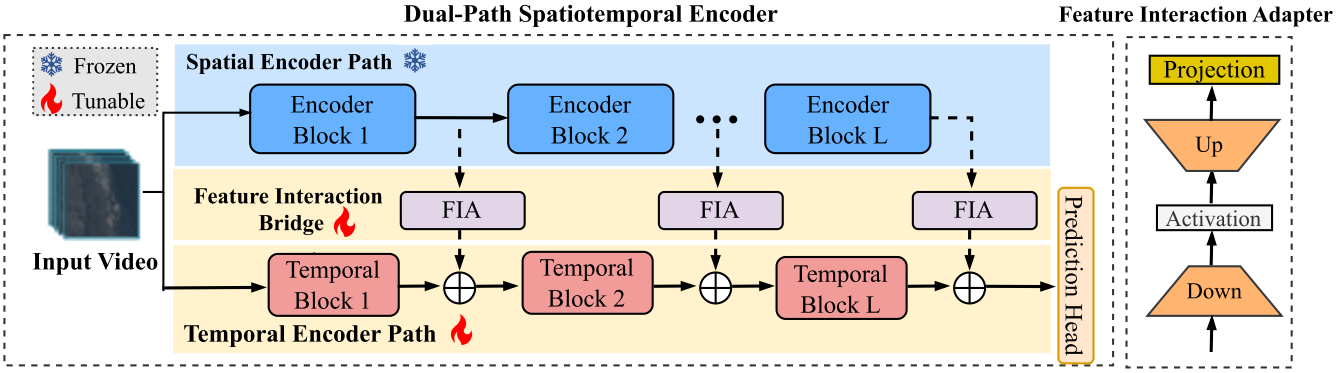


Fig. 2. Architecture of the proposed dual-path spatiotemporal encoder. The spatial path consists of frozen encoder blocks, while the temporal path is tunable and interacts with the spatial features through FIA.

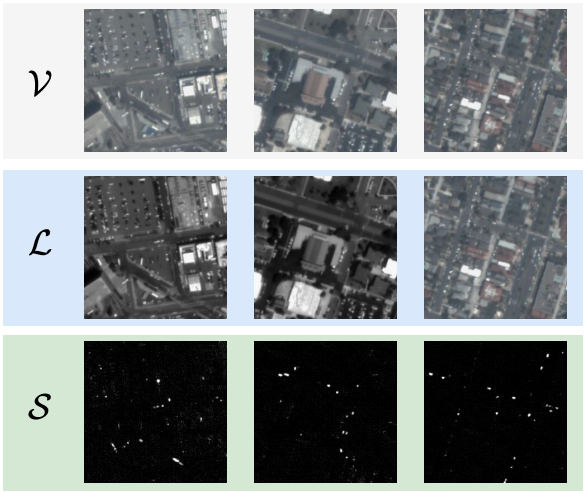


Fig. 3. Illustration of satellite video frames $\mathcal{V}$, their corresponding low-rank background components $\mathcal{L}$, and sparse foreground targets $\mathcal{S}$.

The motion reconstruction branch is designed to recover the temporal dynamics embedded within each masked tube-patch, which is particularly challenging in satellite videos due to sparse moving targets and a low signal-to-noise ratio against a predominantly static background. To provide supervision for motion reconstruction, we follow established methodologies in satellite video processing [48], [49], [50] and leverage tensor robust principal component analysis (TRPCA) [51] to decompose the raw video tensor into low-rank background components, representing static scene elements, and sparse motion components, representing dynamic foreground objects, as illustrated in Fig. 3. Specifically, the TRPCA formulation is given by the following equation:

$$\min_{\mathcal{L},\mathcal{S}} \|\mathcal{L}\|_* + \lambda\|\mathcal{S}\|_1, \quad \text{s.t. } \mathcal{V} = \mathcal{L} + \mathcal{S} \tag{4}$$

where $\|\cdot\|_*$ denotes the tensor nuclear norm, effectively penalizing the rank of the background component $\mathcal{L}$, and $\|\cdot\|_1$ is the ℓ_1 norm applied to the sparse motion tensor $\mathcal{S}$, encouraging sparsity in the motion representation. The parameter λ balances the trade-off between the low-rank and sparse components. The resulting sparse tensor $\mathcal{S}$ serves as the supervision signal for motion reconstruction. TRPCA is performed offline during the construction of the self-supervised pretraining dataset, introducing no additional overhead to the training iterations. Although simpler motion cues, such as frame differencing or optical flow, can be used, we adopt TRPCA to obtain cleaner motion-focused supervision under the highly static background and sparse-motion characteristics of satellite videos. While TRPCA may be less accurate under strong noise, our joint optimization with appearance reconstruction provides a stable complementary signal that reduces sensitivity to imperfect motion information.

For each masked spatiotemporal location $(t, i) \in \Omega$, let $\mathbf{r}_{t,i}$ be the motion values extracted from the sparse component $\mathcal{S}$, and $\hat{\mathbf{r}}_{t,i}$ be the predicted motion values from the motion decoder. To minimize the reconstruction error between these predicted and motion supervision signals, we also use an mse loss

$$\mathcal{L}_{\text{mot}} = \frac{1}{|\Omega|} \sum_{(t,i)\in\Omega} \|\mathbf{r}_{t,i} - \hat{\mathbf{r}}_{t,i}\|_2^2. \tag{5}$$

This motion reconstruction loss guides the motion decoder to capture and reproduce the intricate temporal changes and movements within the satellite video, thereby fostering the learning of discriminative motion-aware features. The final self-supervised loss combines appearance and motion reconstruction

$$\mathcal{L}(\theta_{\text{tm}}, \phi) = \alpha\, \mathcal{L}_{\text{app}} + \beta\, \mathcal{L}_{\text{mot}} \tag{6}$$

where α and β are both set to 0.5, since the two reconstruction losses are mse terms averaged over masked tube tokens and the motion target is consistently normalized, making equal weighting stable without additional tuning. By jointly optimizing both appearance and motion reconstruction, the framework is capable of learning robust spatiotemporal representations essential for SVSU tasks.

B. Spatiotemporal Network With Dual-Path Architecture

Motivated by the data imbalance, where RS imagery is abundant but publicly available satellite video data for SSL remains scarce, training both spatial and temporal encoders from scratch would be inefficient and prone to weak representations. Therefore, we decouple spatial semantics from temporal dynamics in a dual-path design, as shown in Fig. 2. The spatial path is composed of standard Transformer encoder

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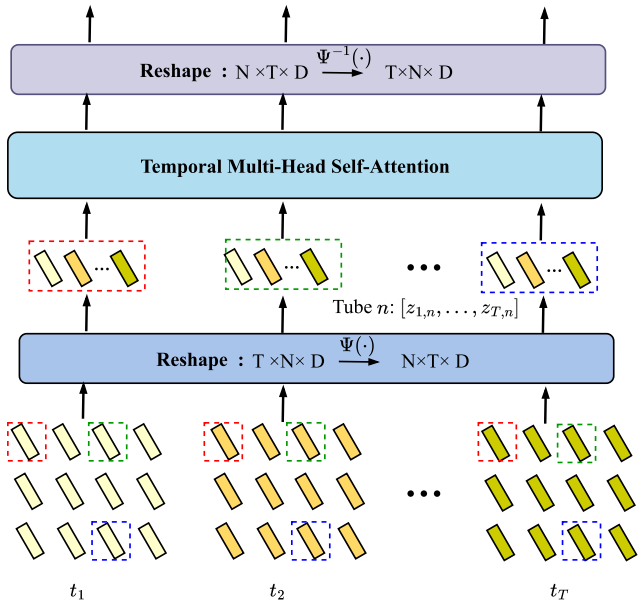


Fig. 4. Illustration of the temporal MHSA module. Tube tokens across different time steps are reshaped and processed by temporal attention to capture dependencies, and then reshaped back to the original format for subsequent processing.

blocks pretrained on large-scale RS images and remains frozen during both pretraining and downstream evaluation. This design choice preserves transferable spatial priors and helps avoid catastrophic forgetting when satellite video data are limited. In parallel, a trainable temporal encoder consisting of temporal attention blocks operating along the time axis is used to learn interframe dependencies and motion cues specific to satellite video. To bridge the frozen spatial encoder and the tunable temporal encoder, a lightweight FIA is introduced, which transforms high-dimensional spatial tokens through a bottleneck MLP and projects them into the temporal feature space. The labels "frozen" and "tunable" in Fig. 2 indicate whether the corresponding modules are fixed or updated during training.

To capture temporal relations, the temporal encoder uses a Transformer that applies multihead self-attention (MHSA) along the time axis. As illustrated in Fig. 4, tokens at the same spatial index across consecutive frames are grouped into temporal tubes, processed by temporal MHSA, and then reshaped back to the original spatiotemporal layout for subsequent fusion. Let $\mathbf{z}_{n,t} \in \mathbb{R}^D$ be the feature at position $n \in \{1, \ldots, N\}$ and frame $t \in \{1, \ldots, T\}$. For each spatial index n, we collect the sequence of tokens over all frames and add a temporal positional embedding

$$\mathbf{Z}_{0,n} = [\mathbf{z}_{n,1}, \mathbf{z}_{n,2}, \ \ldots, \ \mathbf{z}_{n,T}] \tag{7}$$

where the subscript 0 indicates the initial input layer of the temporal encoder.

The temporal encoder is composed of S stacked Transformer blocks that operate only along the temporal axis. At the sth block, the representation is updated as follows:

$$\mathbf{Z}_{s,n} = \mathrm{MLP}\Big(\mathbf{Z}_{s-1,n} + \mathrm{MHSA}(\mathrm{LN}(\mathbf{Z}_{s-1,n}))\Big) \tag{8}$$

for $s = 1, \ldots, S$. Here, MHSA performs multihead self-attention across the T tokens associated with the same spatial position n, thereby capturing both short and long range temporal dependencies.

Freezing the spatial encoder may introduce a distribution mismatch between frozen spatial tokens and trainable temporal tokens. To address this issue, we introduce the FIA module to bridge the frozen spatial encoder and the trainable temporal encoder. The FIA aligns high-dimensional spatial tokens with temporal tokens through a learned transformation rather than simple concatenation, enabling effective knowledge transfer from image-pretrained representations. The FIA is implemented as a lightweight two-layer MLP with a bottleneck, a nonlinear activation, and a final projection, providing a compact yet effective transformation for stable spatiotemporal fusion. Formally, for an input feature vector $\mathbf{h}$ from the spatial encoder, the adapter operation is defined as follows:

$$f_{\mathrm{adp}}(\mathbf{h}) = \mathbf{W}_{\mathrm{proj}}\, \mathbf{W}_{\mathrm{up}}\, \sigma(\mathbf{W}_{\mathrm{down}}\, \mathbf{h}) \tag{9}$$

where $\mathbf{W}_{\mathrm{down}} \in \mathbb{R}^{d_b \times d}$ and $\mathbf{W}_{\mathrm{up}} \in \mathbb{R}^{d \times d_b}$ constitute the bottleneck MLP. Here, d is the dimension of the spatial encoder features, and $d_b \ll d$ is the bottleneck dimension, significantly reducing the parameter count and computational cost. The function $\sigma(\cdot)$ represents a nonlinear activation function, such as GELU, which introduces essential nonlinearity for enhanced feature representation. Finally, $\mathbf{W}_{\mathrm{proj}} \in \mathbb{R}^{d' \times d}$ is a projection matrix that maps the adapted features into the specific dimension d' of the temporal encoder. During self-supervised pretraining, an encoder–decoder architecture with two lightweight Transformer decoders is adopted. Each decoder consists of four Transformer layers with an embedding dimension of 384, and both decoders share the same configuration while reconstructing different targets. After pretraining, both decoders are discarded, and the pretrained encoder is retained for downstream tasks; for SVSU tasks, a linear classifier is attached on top of the encoder representations. Although the spatial backbone is frozen, our self-supervised pretraining optimizes the trainable temporal path and the FIA module to adapt to satellite video distributions and strengthen temporal modeling.

C. Self-Supervised Pretraining Setup

We curate a satellite video dataset for self-supervised pretraining from multiple sources covering diverse regions and scene types, including roads, bridges, intersections, airport runways, shipping channels, and coastal waters. All raw videos and sequential images are converted to a unified AVI format and then cropped with a sliding window to extract spatiotemporal clips at 256 × 256 pixels. Because a large portion of satellite video is static or near-static, we apply an automated EOI detector [24] to prioritize clips likely to contain sustained motion. Although the detector can produce false positives (FPs) and false negatives (FNs), this filtering biases the dataset toward temporally dynamic content. Each selected clip is then resized to 224 × 224 pixels. The clip is standardized to 16 frames by dividing the source sequence into 16 temporal segments and uniformly sampling one frame

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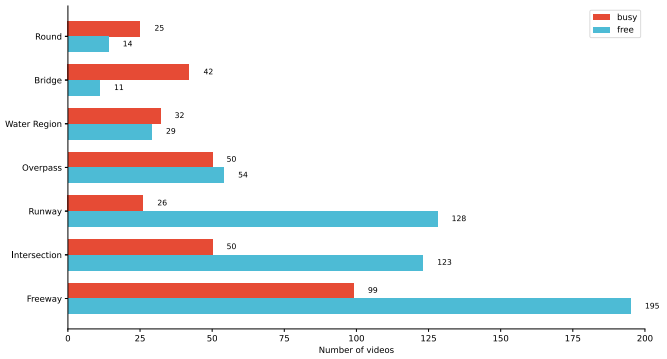


Fig. 5. Number of video samples in the SVSC-14 dataset across 14 scene categories.

from each segment. This pipeline produces about 6500 motion-centric clips for self-supervised pretraining.

The spatial path is a frozen ViT-base initialized with publicly released RS weights [52]; its parameters φ remain fixed. The temporal path and the FIA are trainable and focus on modeling interframe dynamics and fusing with frozen spatial tokens. During pretraining, only θ_{tp} and θ_{adp} are updated, and the objective is

$$\theta^* = \arg\min_{\theta_{\text{tp}},\theta_{\text{adp}}} \mathcal{L}(\mathbf{F};\varphi). \tag{10}$$

We use AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.95$, and weight decay 0.05. The learning rate follows a cosine-decay schedule with a 40-epoch warmup; the base learning rate is 3.75×10^{-5}. The effective batch size is 32, with four GPUs each processing eight video clips. Training runs for 2000 epochs. Data augmentation includes random resized crop, horizontal flip, and color jitter, and all frames are normalized per channel. During pretraining, we adopt a tube masking ratio of 75%, following standard practice in masked video modeling, such as VideoMAE [40].

IV. Experiments

A. Experimental Details

To comprehensively evaluate the effectiveness of the proposed framework, we conduct experiments on the SVSC-14 dataset. The dataset is constructed from Jilin-1 satellite videos with a spatial resolution of 0.92 m. It encompasses 14 scene categories covering diverse traffic and shipping activities. The distribution of samples across all categories is illustrated in Fig. 5. Each scene instance consists of approximately 320 frames with a spatial resolution of 256×256 pixels. The dataset is randomly partitioned into two subsets: 442 video clips are used for training, and 436 clips for testing. The training process adopts the AdamW optimizer with $\beta = (0.9, 0.999)$, a weight decay of 0.15, and cosine decay learning rate scheduling. The initial learning rate is set to 1×10^{-3} with a batch size of 32. A linear warmup is applied during the first four epochs, and the model is trained for a total of 30 epochs. Data augmentation strategies, including random resized cropping, horizontal flipping, and color jittering, are applied to enhance robustness.

TABLE I
Performance Comparison of Different Methods on the SVSC-14 Dataset

Method	OA (mean (std)) (%)	AA (mean (std)) (%)
TANet [53]	61.28 (0.99)	44.12 (1.54)
TIN [54]	71.35 (1.62)	61.31 (1.23)
UniFormerV2 [55]	68.18 (1.42)	57.76 (0.92)
TimeSformer [56]	77.46 (0.92)	67.64 (1.38)
VideoSwin [57]	78.96 (1.04)	68.25 (0.97)
ST-MAE [58]	72.29 (0.72)	66.73 (0.49)
VideoMAE [40]	74.03 (1.02)	68.28 (0.84)
Ours	**81.05 (0.84)**	**71.16 (1.14)**

For comparison, the proposed method is evaluated against representative baselines, including temporal modeling networks, such as TANet [53] and TIN [54], Transformer-based architectures, such as UniFormerV2 [55], TimeSformer [56], and VideoSwin [57], and self-supervised video pretraining models such as ST-MAE [58] and VideoMAE [40]. The experimental performance is measured using overall accuracy (OA) and average accuracy (AA). For all compared methods, the official pretrained checkpoints released by the original authors are loaded; self-supervised pretraining baselines are further pretrained on the same satellite video corpus before fine-tuning on the target dataset, whereas supervised classification baselines are directly fine-tuned on the target dataset.

B. Experimental Results Analysis

Table I presents a comprehensive summary of the quantitative results for SVSU, comparing our proposed framework against a range of contemporary methods. The results are reported as mean $\pm$ standard deviation over ten runs while keeping the same SVSC-14 data split and training protocol. Methods that integrate temporal information through convolutions, such as TANet and TIN, yield lower accuracy, reflecting the limitations of local temporal aggregation when modeling the long-range and often sparse motion patterns in satellite video. Transformer-based models perform better by leveraging global attention. TimeSformer, for instance, enhances sensitivity to both appearance and motion by explicitly separating spatial and temporal attention. VideoSwin, with its innovative shifted-window attention and hierarchical design, proves particularly effective. Its ability to capture multiscale structures in both space and time is well-suited for complex satellite scenes that demand not only broad spatial context but also coupled short-term and long-term temporal cues. Self-supervised masked reconstruction approaches, including VideoMAE and ST-MAE, achieve mid-level performance; lacking explicit temporal modeling or a multiscale hierarchy, they do not fully exploit motion cues in satellite videos. In contrast, under limited labeled supervision, the proposed framework attains the highest accuracy by leveraging self-supervised pretraining on a curated satellite video corpus to enhance temporal modeling while keeping the RS-image pretrained spatial ViT frozen and updating only the temporal path and the FIA module to mitigate overfitting. The resulting

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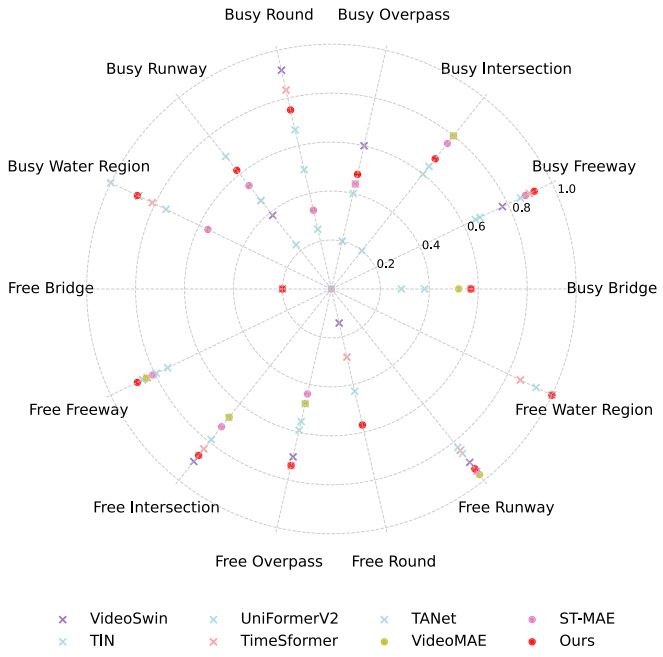


Fig. 6. Per-category classification accuracy comparison among different methods on the SVSC-14 dataset.

gains are consistent with the dual-path design that decouples spatial semantics from temporal dynamics and strengthens spatiotemporal representations. However, since both spatial and temporal encoders are implemented using plain ViT backbones without additional architectural complexity, the improvements are primarily attributed to the proposed formulation and training strategy rather than specialized encoder engineering.

Fig. 6 provides a detailed per-category classification accuracy comparison on the SVSC-14 dataset, offering insights into the nuanced performance of various methods across different scene types. Our proposed approach consistently forms the outer envelope, achieving the highest performance across the majority of scene categories, thereby indicating a clear advantage over alternative methods. This is particularly significant considering that operational satellite video scenes often exhibit a markedly imbalanced class distribution, with pronounced differences in the frequency of various scene types. The SVSC-14 dataset reflects this reality, making robust performance across diverse categories a crucial indicator of a model's practical utility. The high accuracy achieved by our method across both high- and low-frequency classes strongly suggests that the model is resilient to class imbalance and possesses strong generalization capabilities, rather than merely overfitting to more common scene types. This observed robustness and consistent performance across categories directly stem from our unique architectural design and learning strategy. First, the frozen image-pretrained ViT backbone plays a pivotal role by supplying stable spatial priors. This design choice prevents the feature space from undesirably drifting toward representations dominated by frequent categories, thereby ensuring a foundational understanding for all scene types. Second, the temporal path, meticulously optimized through our self-supervised objectives, is specifically designed to emphasize and capture crucial motion persistence and magnitude signals. These dynamic cues are inherently less tied to specific class frequencies and demonstrate high transferability across various scenes, providing discriminative information that benefits even rare categories.

Fig. 7 visually illustrates the classification confusion matrices for various methods on the SVSC-14 dataset, providing a granular view of their discriminative capabilities. Our proposed method consistently exhibits crisper diagonal entries and significantly fewer off-diagonal entries across the majority of categories. The most profound impact of our method is evident in its ability to differentiate between scene pairs that share a high degree of visual similarity in their static appearance but diverge significantly in their dynamic states. For instance, the misclassification rates between the busy and free variants of rounds and runways are notably reduced. The proposed dual-path architecture, together with motion-aware self-supervision based on TRPCA, enables the model to capture subtle interframe dynamics that distinguish visually similar but dynamically different states even when motion cues are weak.

C. Ablation Study

A comprehensive ablation study is conducted to rigorously validate the effectiveness of each architectural design choice. The results are summarized in Table II. The baseline configuration, which utilizes only a fine-tuned spatial encoder, achieves an initial level of performance. The introduction of a trainable temporal encoder alongside the fine-tuned spatial one results in a substantial performance gain, highlighting the critical necessity of explicitly modeling temporal dynamics. The third configuration reveals the benefit of preserving pretrained knowledge. By freezing the spatial encoder instead of fine-tuning it, performance further improves. This result strongly supports our hypothesis that preventing catastrophic forgetting and leveraging robust spatial priors from large-scale image pretraining is more effective than retraining the spatial backbone on limited satellite video data. In contrast, fully fine-tuning the spatial encoder may offer stronger domain adaptation when abundant data are available; however, in practical satellite video scenarios where labeled data are limited, this strategy becomes less stable and often leads to degraded performance. The integration of the FIA provides the final performance boost, leading to our top-performing model. This demonstrates the FIA's vital role in effectively aligning and fusing features from the distinct spatial and temporal pathways.

Table III examines how different self-supervised reconstruction objectives affect pretraining. As a reference, we consider a baseline in which the trainable subset, namely the temporal path and the FIA, starts from random initialization, while the spatial path remains frozen with image-pretrained weights; this setting reflects having only the spatial prior and no video-specific pretraining. Pretraining with the appearance reconstruction objective alone consistently improves performance, indicating that reconstructing masked content strengthens spatial context and provides effective supervisory signals for the interaction between the frozen and trainable

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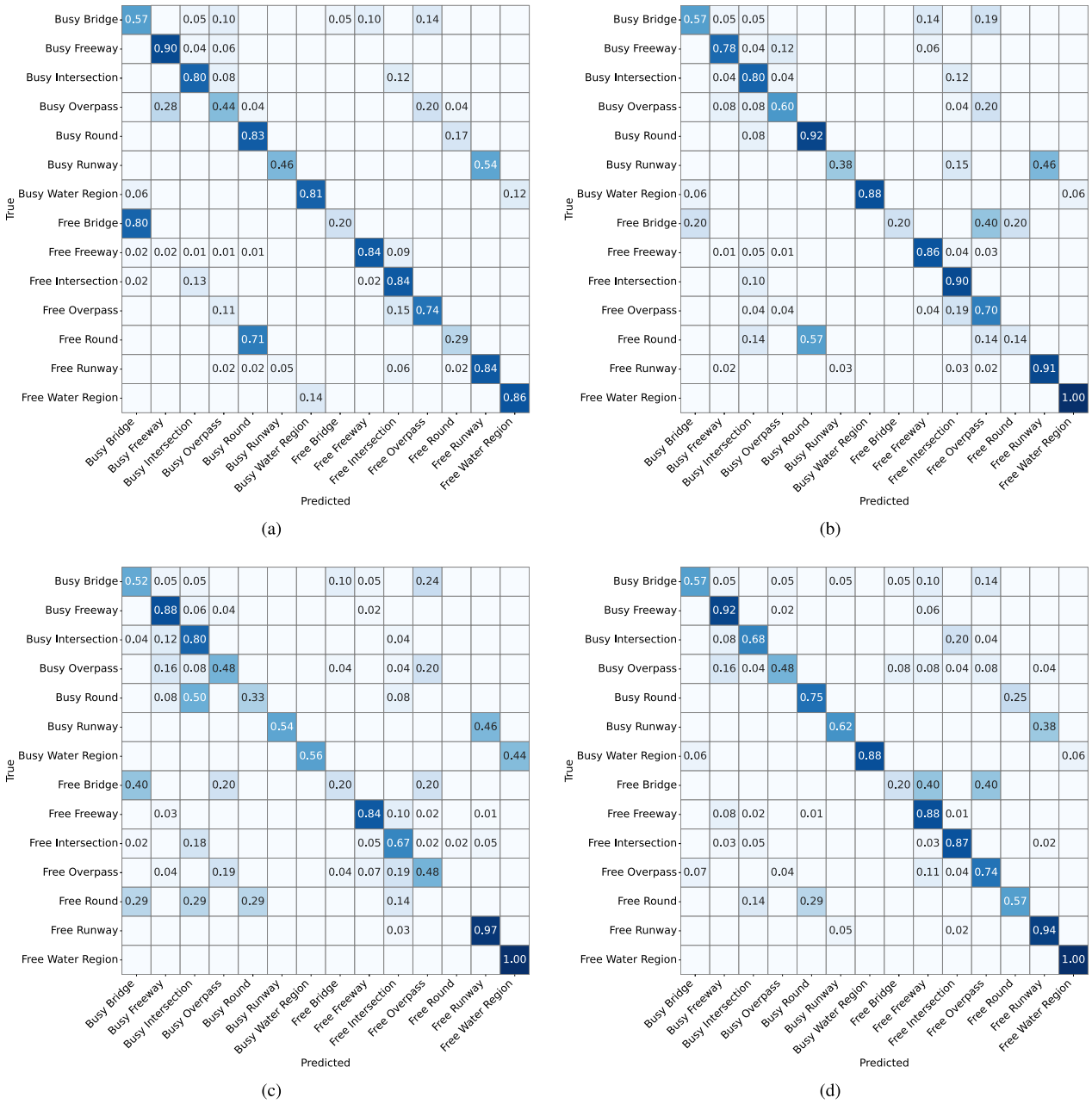


Fig. 7. Classification confusion matrices for different methods on the SVSC-14 dataset. (a) TimeSformer. (b) VideoSwin. (c) VideoMAE. (d) Our method. Darker diagonal cells indicate more accurate per-class predictions.

TABLE II

ABLATION STUDY ON ARCHITECTURAL CONFIGURATIONS. THE CHECKMARK (✓) INDICATES THE COMPONENT IS INCLUDED, WITH ITS STATE (FT FOR FINE-TUNED, FRZ FOR FROZEN) SPECIFIED

Components			Performance	
Spatial	Temporal	FIA	OA (%)	AA (%)
✓(*ft*)	–	–	61.14	48.47
✓(*ft*)	✓(*ft*)	–	78.54	69.44
✓(*frz*)	✓(*ft*)	–	80.51	70.84
✓(*frz*)	✓(*ft*)	✓(*ft*)	**81.19**	**72.03**

TABLE III

ABLATION STUDY ON THE IMPACT OF SELF-SUPERVISED RECONSTRUCTION OBJECTIVES DURING PRETRAINING

Pretraining Objectives			Performance	
Pretraining	Appearance	Motion	OA (%)	AA (%)
–	–	–	77.14	67.68
✓	✓	–	79.72	69.74
✓	–	✓	78.86	68.76
✓	✓	✓	**81.19**	**72.03**

paths. Using only the motion reconstruction objective also yields clear gains over the baseline, confirming that sensitivity to temporal persistence and motion magnitude is critical for discriminating scene states. The strongest and most stable

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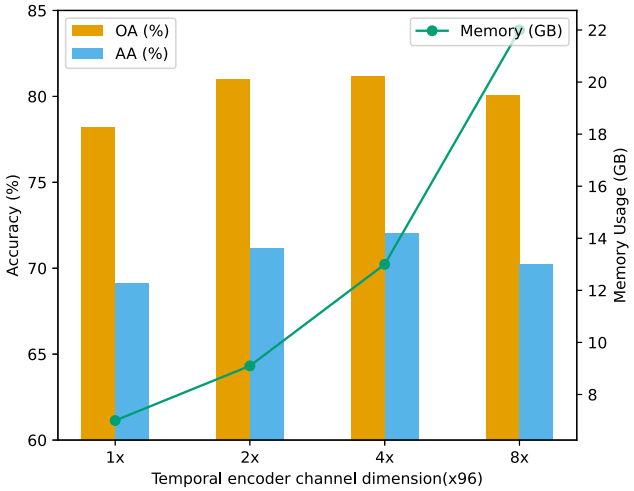


Fig. 8. Impact of temporal encoder channel dimension on classification accuracy and memory usage on the SVSC-14 dataset.

TABLE IV
SENSITIVITY TO THE NUMBER OF LABELED TRAINING CLIPS ON THE SVSC-14 DATASET

Training clips	OA (%)	AA (%)
300	77.29	66.42
200	71.33	62.72
100	66.74	56.32

results arise when the two objectives are combined: appearance reconstruction preserves reliable spatial semantics, whereas motion reconstruction emphasizes dynamic cues, and their complementarity produces more balanced per-category behavior and sharper decision boundaries. All compared baselines in Table I are fine-tuned on SVSC-14 using the same data split and the same downstream training protocol, while ablation studies are conducted to further isolate the performance gains attributable to the proposed dual-path architecture and self-supervised objectives.

Fig. 8 presents a crucial analysis of the temporal encoder's channel dimension and its impact on classification accuracy and memory footprint on the SVSC-14 dataset. Initially, increasing the channel dimension from its smallest setting yields noticeable accuracy gains. However, beyond a moderate width, the curve enters a regime of diminishing returns: accuracy changes only marginally while memory consumption continues to rise. When the channel dimension becomes excessively large, accuracy declines slightly as memory cost keeps growing, suggesting redundancy in the learned features and increased susceptibility to overfitting, particularly under class imbalance. Based on this analysis, the temporal encoder width is set to 384 as a trade-off between accuracy and memory, and the same setting is kept unchanged when transferring the framework from SVSC-14 classification to EOI detection, indicating reasonable robustness across SVSU tasks.

To analyze sensitivity to labeled data size, the pretrained model is fine-tuned using progressively smaller subsets of the SVSC-14 training set. Each subset is constructed by uniformly subsampling clips from all categories to preserve the original class distribution, such that the observed performance variations are primarily attributed to the amount of supervision rather than class imbalance. The corresponding results are summarized in Table IV, which indicates that the proposed framework remains reasonably label-efficient even under limited supervision. We also note that broader cross-dataset and cross-region generalization under domain shifts remains an important direction for future work.

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE EOI DETECTION DATASET

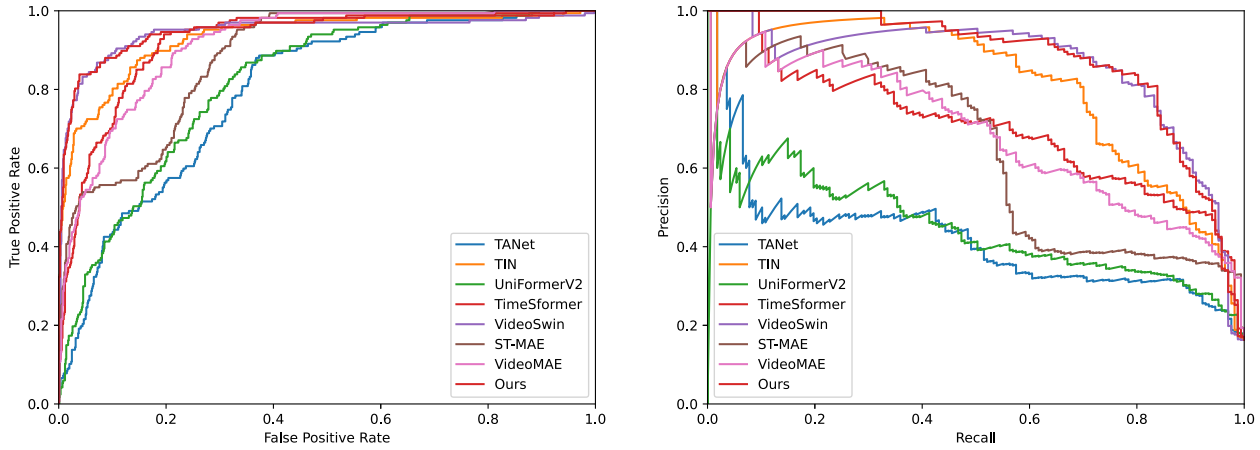
Method	AUC	AP
TANet [53]	0.796	0.419
TIN [54]	0.933	0.805
UniFormerV2 [55]	0.818	0.445
TimeSformer [56]	0.922	0.703
VideoSwin [57]	0.947	0.858
ST-MAE [58]	0.878	0.641
VideoMAE [40]	0.913	0.682
Ours	**0.953**	**0.875**

D. Discussion

In addition to evaluation on SVSC-14 as a standardized benchmark for SVSU, the generalization capability of the proposed framework is further assessed by extending it to the task of EOI detection in satellite videos. According to the established protocol, EOI instances are defined as spatiotemporal blocks containing dynamic targets, whereas non-EOI instances correspond to static scenes without dynamic activities. The training set consists of 900 samples, including 310 EOI and 590 non-EOI instances, while the test set comprises 1000 samples, with 160 EOI and 840 non-EOI instances. Each instance covers a spatial area of 128×128 pixels and contains 300 frames. We evaluate EOI detection using the receiver operating characteristic (ROC) curve with its area under the curve (AUC), and the precision–recall (PR) curve with average precision (AP). Scores are converted to labels by sweeping a decision threshold τ: a sample is positive if $s \geq \tau$ and negative otherwise, which yields counts of true positives (TPs), FP, true negatives (TNs), and FN. The ROC curve plots the TP rate $\mathrm{TPR} = (\mathrm{TP}/(\mathrm{TP} + \mathrm{FN}))$ against the FP rate $\mathrm{FPR} = (\mathrm{FP}/(\mathrm{FP} + \mathrm{TN}))$ as τ varies, and the AUC is the area under this curve, providing a single measure of ranking quality. For imbalanced data, which is common in satellite video EOI detection, the PR curve is more diagnostic: it traces $\mathrm{Precision} = (\mathrm{TP}/(\mathrm{TP} + \mathrm{FP}))$ versus $\mathrm{Recall} = (\mathrm{TP}/(\mathrm{TP} + \mathrm{FN}))$ across thresholds and emphasizes performance on the positive class. AP summarizes the PR curve as the area under it, computed from interpolated precision values over increasing recall. All other training configurations remain identical to those described in the previous experiments.

Table V reports the performance comparison of various methods on the EOI dataset, evaluated in terms of AUC and AP. It is observed that Transformer-based methods

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(a) (b)

Fig. 9. ROC and PR curves of different methods for the EOI dataset. (a) ROC curves. (b) PR curves.

generally leverage attention mechanisms to yield stronger results. Among these, VideoSwin shows a particularly strong performance, likely due to its windowing mechanism that effectively models both short-term and long-term motion information and inherently possesses multiscale capabilities. In contrast, self-supervised pretraining methods that do not explicitly model temporal motion dynamics, such as ST-MAE and VideoMAE, exhibit comparatively limited effectiveness for EOI detection, a task highly sensitive to subtle temporal motion cues. Our proposed framework consistently achieves the highest values across both AUC and AP, demonstrating superior performance compared to all baseline approaches.

Fig. 9 presents the ROC and PR curves of different methods on the EOI dataset. The proposed method consistently achieves higher TP rates across a wide range of FP rates in the ROC curve, as well as superior precision over varying recall levels in the PR curve. These results confirm that the proposed framework not only improves overall detection metrics but also maintains stable and robust performance across different decision thresholds. This leading performance stems from our dual-path structural design and the specialized pretraining of features for motion information. Notably, even though both our temporal and spatial branches employ pure ViT architectures without explicit multiscale design, the framework still demonstrates superior performance. This highlights the effectiveness of our design in capturing critical spatiotemporal features for the EOI detection task.

V. Conclusion

In this article, we present a novel dual-path self-supervised spatiotemporal learning framework for SVSU. The framework disentangles spatial and temporal representation learning, leverages a frozen image-pretrained ViT to preserve strong spatial priors, and incorporates a temporal encoder to capture motion dynamics unique to satellite videos. The FIA module further enables effective fusion between the two paths. To empower spatiotemporal modeling, we design a dual reconstruction self-supervised objective that jointly optimizes masked spatial appearance recovery and motion reconstruction. This strategy compels the temporal branch to learn discriminative motion-aware features while maintaining the semantic consistency of the frozen spatial path. Extensive experiments on the SVSC-14 dataset show that our framework consistently outperforms comparative methods. Evaluations on the EOI detection task further verify the robustness and scalability of the proposed paradigm. Ablation studies confirm the effectiveness of the proposed framework.

For future work, we will further develop this framework along two directions. First, the proposed framework can be extended to more advanced and efficient encoders with multiscale modeling and longer temporal receptive fields, while maintaining the decoupled design of spatial and temporal paths. Second, we aim to extend and evaluate the framework on a wider spectrum of SVSU tasks beyond scene classification and EOI detection, thereby assessing the generality and scalability of the proposed paradigm in broader satellite video scenarios.

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